

Infinite Communication Complexity and KW Games

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Abstract

We characterize the Borel sets with an infinite version of Karchmer and Wigderson’s game linking finite circuit complexity to communication complexity ([6]). To this end, we formulate an infinite version of communication complexity and prove that a given subset of the Cantor space can be computed by an infinite circuit if and only if a certain infinite communication game is solvable. This, combined with the correspondence between infinite circuits and Borel sets outlined by Sipser in [9], gives us the characterization. We utilize this connection to provide new elementary and purely combinatorial proofs of some classical results in descriptive set theory, including the analytic separation theorem and the equivalence of monotone and positive Borel sets. Another consequence is a characterization of Borel separability via the winner of a certain “cut-and-choose” game, which we use to obtain new proofs of the fact that no infinite parity function is Borel and the separation of the Borel sets from the analytic sets.

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1 Introduction

Karchmer-Wigderson (KW) games ([6]) characterize the minimum circuit depth needed to compute a boolean function in terms of a communication game. This connection allows us to prove elusive circuit lower bounds by transferring the problem to the field of communication complexity, where the state of lower bounds is far better. KW games have been used to prove several state-of-the-art lower bounds on explicit functions, such as the monotone circuit depth of perfect matching ([8]), and they remain a promising technique for resolving major open problems in complexity theory such as the NC^1 vs. P question ([5]).

In this paper, we generalize communication complexity and KW games to the infinite setting. We extend Karchmer and Wigderson’s results to characterize sets computable by infinite circuits in terms of infinite KW games. Combined with a connection between Borel sets and infinite circuits ([9]), this gives us a complete characterization of Borel sets in terms of KW games. We then use this correspondence in both directions to prove positive and negative results about Borel sets.

Though the applications of KW games we discuss are not new results, the proofs are purely combinatorial and quite elementary. In particular, we do not rely on any external machinery from descriptive set theory. Many of the results we prove have major open problems regarding P, NP, and coNP as finite analogs. By finding purely combinatorial proofs of the infinite versions of these claims, we can hopefully develop ideas helpful in eventually resolving these questions.

1.1 Outline and summary of results

We discuss preliminaries in Section 2, focusing on the connection between Borel/analytic sets and infinite deterministic/nondeterministic circuits. Several of the basic results such as Theorem 2.12 are interesting in their own right, especially for those less familiar with circuit complexity. We also provide an overview of finite communication complexity in Section 2.5, which includes all background needed for this paper.

Our new contributions begin in Section 3, where we introduce and motivate a model of infinite communication complexity. We show how techniques from the finite setting generalize seamlessly to this new framework. In Section 4, we formulate an infinite analog of Karchmer-Wigderson (KW) games ([6]). We use these games to provide a new characterization of the Borel sets, as well as the Borel sets computable by monotone circuits (i.e. the *positive* Borel sets, see [7]), in terms of infinite communication complexity.

In Section 5, we use KW games to provide new elementary and purely combinatorial proofs of the analytic separation theorem (Theorem 5.1) and the fact that a Borel set is monotone if and only if it is computable by a monotone circuit (Theorem 5.2). In Section 6, we prove impossibility results using KW games, first explaining why many techniques from finite communication complexity fail to prove lower bounds. We circumvent this obstruction by using KW games to prove the equivalence of Borel separability and a winning strategy in a certain cut-and-choose game, defined in Section 6.2. We then use this game to provide

proofs that no infinite parity function is Borel (Theorem 6.10) and that the set of ill-founded trees is not Borel (Theorem 6.14). We conclude by discussing the difficulties of translating these proofs to the finite setting in Section 7.

2 Preliminaries

Let $\Sigma = \{0, 1\}$ denote the binary alphabet, and set $\Sigma^* = \bigcup_{n=0}^{\infty} \Sigma^n$. Let $a_1 \cdots a_k b$ denote the concatenation of strings $a_1, \dots, a_k \in \Sigma^*$ and $b \in \Sigma^* \cup \Sigma^\omega$. We consider the *Cantor space* Σ^ω , equipped with the topology generated by basic open sets of the form $x\Sigma^\omega$ for $x \in \Sigma^*$.

2.1 Borel and analytic sets

Definition 2.1. We call a set Σ_0 or Π_0 if it is clopen. A set is Σ_α (resp. Π_α) if it is a countable union (resp. intersection) of sets $S_1, S_2, \dots$, each of which is in Π_β (resp. Σ_β) for some $\beta < \alpha$ (note that β may depend on the set S_i).

This definition agrees with the usual definition of the Borel hierarchy since the Cantor space is zero-dimensional (has a clopen basis). In particular, the open (resp. closed) sets of Σ^ω are precisely the countable unions (resp. intersections) of clopen sets.

Definition 2.2. A set $B \subseteq \Sigma^\omega$ is *Borel* if it is Σ_α for some $\alpha < \omega_1$. The *rank* of a Borel set B is the smallest α such that $B \in \Sigma_\alpha \cup \Pi_\alpha$.

Definition 2.3. A set $A \subseteq \Sigma^\omega$ is *analytic* if it is the projection of a Borel set; i.e. if there exists a Borel set $B \subseteq \Sigma^\omega \times \Sigma^\omega \cong \Sigma^\omega$ such that

$$A = \{x \in \Sigma^\omega : (x, y) \in B \text{ for some } y \in \Sigma^\omega\}.$$

2.2 Infinite circuits

A *circuit* C on variables $x_1, x_2, \dots$ is a countable, directed, well-founded graph with vertex set consisting of literals $0, 1, x_1, \bar{x}_1, x_2, \bar{x}_2, \dots$, as well as $\wedge$ - and $\vee$ -gates with countable fan-in. We call these vertices *gates*. Given a setting of the x_i 's to 0 or 1, each gate G in C evaluates to 0 or 1 in the obvious manner (the existence and uniqueness of this evaluation is guaranteed by the well-foundedness of C). We write $G(x)$ to denote this evaluation. One vertex G_{out} of C is labeled as the *output*, and we write $C(x) = G_{\text{out}}(x)$. Thus C computes a function from Σ^ω to Σ , and we say that C computes the set $C^{-1}(\{1\})$.

Remark 2.4. Allowing for $\neg$ -gates would not increase the power of these circuits, since De Morgan's law allows us to propagate all negations to the bottom layer of the circuit. We can also assume without loss of generality that gates are alternating (i.e. no $\vee$ -gate has another $\vee$ -gate as a child and no $\wedge$ -gate has another $\wedge$ -gate as a child), since we can merge any non-alternating gates. This second observation justifies the following definition:

Definition 2.5. We call a circuit Σ_0 or Π_0 if it is finite. A Σ_α -circuit (resp. Π_α -circuit) is an $\vee$ -gate (resp. $\wedge$ -gate) with countably many children $C_1, C_2, \dots$, each of which is a Π_β -circuit (resp. Σ_β -circuit) for some $\beta < \alpha$ (which can depend on the circuit C_i).

Definition 2.6. The *rank* of a circuit C is the least ordinal α such that C is a Σ_α - or Π_α -circuit.

Remark 2.7. By inductively duplicating shared subcircuits, we can modify any circuit so that all gates have out-degree 1 without increasing the circuit's rank. This, along with the fact that any countable well-founded tree has rank less than ω_1 , tells us that the rank of any circuit is less than ω_1 after merging non-alternating gates.

2.3 Equivalence of Borel sets and circuits

Our similar choice of notation between Borel sets and circuits is deliberate, as these notions correspond precisely. The following result and its corollaries allow us to describe Borel subsets of Σ^ω and functions $f: \Sigma^\omega \rightarrow \Sigma$ using circuits.

Lemma 2.8. *A set $S \subseteq \Sigma^\omega$ is clopen if and only if it is computed by a finite circuit.*

Proof. Any open set $S \subseteq \Sigma^\omega$ can be written as a union of basic open sets. If S is also closed, it is compact as a closed subset of the compact space Σ^ω . Thus any clopen set S can be written as a finite union of basic open sets. Each of these basic open sets is computed by a finite circuit, and S is computed by the finite disjunction of these circuits.

Conversely, any circuit consisting of a single literal computes a set of the form $\{x \in \Sigma^\omega : x_i = b\}$ for some $i \in \mathbb{N}$ and $b \in \Sigma$. This set is clopen as the finite union of the clopen sets $\bigcup_{w \in \Sigma^{i-1}} wb\Sigma^\omega$. As the clopen sets are closed under finite unions and intersections, this means that any set S computed by a finite circuit is also clopen. $\square$

Since the basic open sets $x\Sigma^\omega$ generating the topology on Σ^ω are clopen, the open sets in Σ^ω are precisely the countable unions of clopen sets. This fact, along with a simple induction on α , gives us the following:

Corollary 2.9. *For $\alpha \geq 1$, a set $S \subseteq \Sigma^\omega$ is Σ_α (resp. Π_α) if and only if it is computed by a Σ_α (resp. Π_α)-circuit.*

Corollary 2.10. *A subset of Σ^ω is Borel if and only if it is computed by a circuit.*

2.3.1 Borel and tree ranks

The typical notion of ordinal rank for well-founded trees gives us a measure of complexity for circuits, but this measure disagrees with our definition of circuit rank. In particular, all finite circuits have rank 0 by our definition, while the ordinal ranks of the underlying trees can be larger. We can resolve this discrepancy using a modified definition of rank for trees:

Definition 2.11. If T is a well-founded tree, we define the *essential rank* $\text{rank}(v)$ of each node $v \in T$ inductively as follows: If the subtree of T rooted at v is finite, we set $\text{rank}(v) = 0$. Otherwise, we define

$$\text{rank}(v) = \sup\{\text{rank}(w) + 1 : w \text{ is a child of } v\}$$

We define $\text{rank}(T)$ to be the essential rank of the root node of T .

This definition ensures that a tree has rank 0 if and only if it is finite, and induction tells us that the rank of a circuit is precisely the essential rank of its underlying tree.

2.4 Nondeterminism

Just as in the finite setting, we can consider *nondeterministic circuits* which take as input two sets of variables $x_1, x_2, \dots$ and $y_1, y_2, \dots$. We call $y_1, y_2, \dots$ the nondeterministic inputs, and we say that a nondeterministic circuit C computes the set

$$\{x \in \Sigma^\omega : C(x, y) = 1 \text{ for some } y \in \Sigma^\omega\}.$$

It is immediate from Corollary 2.10 that the analytic sets are precisely the sets computed by nondeterministic circuits. In fact, we can make the stronger claim that every analytic set is computed by a depth-2 nondeterministic circuit. The argument works just as in the proof that 3SAT is NP-complete, nondeterministically guessing the values of all gates in the circuit and verifying that the resulting computation is valid. Since this result will be of crucial importance later, we include a proof for completeness:

Theorem 2.12. *A set $A \subseteq \Sigma^\omega$ is analytic if and only if it is computed by a nondeterministic CNF (i.e. a depth-2 nondeterministic circuit whose output gate is an $\wedge$ -gate).*

Proof. The reverse direction is immediate. For the forward direction, let C be a nondeterministic circuit computing A . Let $G_1, G_2, \dots$ be an enumeration of the gates of C , with G_1 being the output gate. We define new nondeterministic variables z_n for $n \in \mathbb{N}$. Then since C defines a monotone function on the literals $0, 1, x_1, \bar{x}_1, \dots, y_1, \bar{y}_1, \dots$, we have $x \in A$ if and only if there exist $y, z \in \Sigma^\omega$ such that $z_1 = 1$ and the following hold for each $n \in \mathbb{N}$:

- If G_n is an $\vee$ -gate with inputs $w_1, w_2, \dots$ (each w_k is some $x_i, \bar{x}_i, y_i, \bar{y}_i$, or z_i for $i \in \mathbb{N}$), then we have $\bar{z}_n \vee \bigvee_{k \in \mathbb{N}} w_k$.
- If G_n is an $\wedge$ -gate with inputs $w_1, w_2, \dots$, then we have $\bigwedge_{k \in \mathbb{N}} (\bar{z}_n \vee w_k)$.

By taking the conjunction of z_1 and all the clauses described above for each gate G_n , we obtain a nondeterministic circuit computing A . After merging non-alternating gates, this circuit becomes our desired CNF. $\square$

Remark 2.13. This result can be rephrased as saying that every analytic subset of Σ^ω is the projection of a G_δ (i.e. Π_2)-subset of $\Sigma^\omega \times \Sigma^\omega$. However, our combinatorial proof is quite different from the standard topological argument (and perhaps more elementary).

2.5 Finite communication complexity

The field of *communication complexity* studies the following problem: Suppose Alice is given a string $x \in X$ and Bob is given a string $y \in Y$. How many bits do they need to communicate in order to compute a given function $f: X \times Y \rightarrow Z$? This question is formalized using the concept of a *protocol*, which is a finite binary tree with internal nodes belonging to either Alice or Bob. Internal nodes v are labeled either with functions $a_v: X \rightarrow \Sigma$ (for Alice's nodes) or $b_v: Y \rightarrow \Sigma$ (for Bob's nodes) describing what Alice or Bob chooses to communicate at that step. These functions can be arbitrary, reflecting the fact that Alice and Bob have unbounded computational power. Leaf nodes are labeled with elements of Z , denoting outputs of the protocol.

A protocol P computes a function $P: X \times Y \rightarrow Z$, where one starts at the root node and traverses down the tree, computing $a_v(x)$ or $b_v(y)$ at each internal node v to determine which child to progress to (the left child if a_v or b_v evaluates to 0, and the right child otherwise), and outputting the value of the leaf node eventually reached. The *cost* of a protocol is the height of the tree. One unfamiliar with communication complexity should verify that these protocol trees correspond precisely to strategies for Alice and Bob, with the cost of the protocol corresponding to the number of bits communicated.

Definition 2.14. The *communication complexity* of a function $f: X \times Y \rightarrow Z$, denoted $D(f)$, is the minimum cost of a protocol computing f .

2.5.1 Fooling sets and lower bounds

The primary goal of communication complexity is to prove lower bounds on $D(f)$. Typically, this is done using the following definitions and theorem:

Definition 2.15. A *rectangle* in $X \times Y$ is a set of the form $A \times B$ for $A \subseteq X$ and $B \subseteq Y$. We call a rectangle R *f-monochromatic* if f is constant on R . The *partition number* $\chi(f)$ of a function f is the smallest number of *f-monochromatic* rectangles needed to partition $X \times Y$.

Theorem 2.16. We have $D(f) \geq \lceil \log_2 \chi(f) \rceil$.

Proof. Suppose f has a protocol P of cost c . For each leaf node v of P , there are functions $a_1, \dots, a_k, b_1, \dots, b_\ell$ and bits $q_1, \dots, q_k, r_1, \dots, r_\ell$ such that the protocol reaches v on input $(x, y) \in X \times Y$ if and only if $x \in a_i^{-1}(\{q_i\})$ for each $1 \leq i \leq k$ and $y \in b_j^{-1}(\{r_j\})$ for each $1 \leq j \leq \ell$. Thus P is constant on the rectangle

$$R_v = \left(\bigcap_{i=1}^k a_i^{-1}(\{q_i\}) \right) \times \left(\bigcap_{j=1}^{\ell} b_j^{-1}(\{r_j\}) \right) \subseteq X \times Y.$$

The rectangles R_v for all leaf nodes v of P partition $X \times Y$, and since P computes f , these rectangles are *f-monochromatic*. There are at most 2^c such rectangles, and this implies that $2^{D(f)} \geq \chi(f)$. The claim follows from taking logarithms. $\square$

We can illustrate this technique by proving sharp lower bounds on the following two functions:

Definition 2.17. We define for $n \in \mathbb{N}$ the function $\text{EQ}_n : \Sigma^n \times \Sigma^n \rightarrow \{0, 1\}$ by $\text{EQ}_n(x, y) = 1$ if $x = y$ and $\text{EQ}_n(x, y) = 0$ otherwise. Similarly, we define $\text{DISJ}_n(x, y) = 1$ if and only if $x_i \wedge y_i = 0$ for all $1 \leq i \leq n$.

Corollary 2.18. We have $D(\text{EQ}_n) = D(\text{DISJ}_n) = n + 1$.

Proof. The upper bound follows from the fact that Alice can use n bits to share x with Bob, and Bob can then compute $\text{EQ}_n(x, y)$ or $\text{DISJ}_n(x, y)$ and use a single bit of communication to branch to the correct leaf node. For the lower bound on $D(\text{EQ}_n)$, consider the set $\mathcal{F}_n = \{(x, x) : x \in \Sigma^n\}$ (we call this set a *fooling set*). If any rectangle $R \subseteq X \times Y$ contains two distinct elements $(x, x), (x', x') \in \mathcal{F}_n \subseteq \text{EQ}_n^{-1}(\{1\})$, then R must also contain the point $(x, x') \in \text{EQ}_n^{-1}(\{0\})$. Thus R is not EQ_n -monochromatic, and the contrapositive tells us that any EQ_n -monochromatic rectangle can contain at most one element of $\mathcal{F}_n$. As $|\mathcal{F}_n| = 2^n$, this means that 2^n EQ_n -monochromatic rectangles are needed to cover $\text{EQ}_n^{-1}(\{1\})$. At least one additional rectangle is needed to cover $\text{EQ}_n^{-1}(\{0\})$, so

$$D(\text{EQ}_n) \geq \lceil \log_2 \chi(\text{EQ}_n) \rceil \geq \lceil \log_2(2^n + 1) \rceil = n + 1.$$

Finally, the lower bound on $D(\text{DISJ}_n)$ follows from the collection $\mathcal{F}_n = \{(x, \bar{x}) : x \in \Sigma^n\}$ also being a fooling set with size 2^n (and from $\text{DISJ}_n^{-1}(\{0\})$ again being nonempty). $\square$

3 Infinite communication complexity

In this section, we aim to define the communication complexity of functions from $\Sigma^\omega \times \Sigma^\omega$ to $\{0, 1\}$ or $\mathbb{N}$. One immediate observation is that our protocol trees must be infinite in order to study any interesting problems. In order to ensure that every protocol computes a well-defined function, we also require that protocols be well-founded. By König's lemma, any well-founded tree which is finitely branching must itself be finite. For this reason, we allow protocol trees to have countably infinite branching. In other words, Alice and Bob communicate by sending natural numbers instead of bits.

Definition 3.1 (Infinite communication protocols). Let $X = Y = \Sigma^\omega$ and Z be at most countable. A *protocol* P for a communication problem $f : X \times Y \rightarrow Z$ is a well-founded, countably branching tree with internal nodes v labeled by functions $a_v : X \rightarrow \mathbb{N}$ or $b_v : Y \rightarrow \mathbb{N}$ and leaf nodes labeled by elements of Z . The function P computes is defined analogously to the finite case.

In the finite setting, every problem has a communication protocol (Alice or Bob can just send the other person their string in its entirety). In the infinite setting, on the other hand, this is not the case (we will prove this shortly). While the cost of a communication protocol is an essential notion in the finite case, the most interesting question in the infinite setting is often whether a protocol even exists. Nonetheless, we can still define the cost of protocols

and communication complexity of functions in the infinite setting. Essential rank is used in place of the ordinary notion of rank so that Corollary 4.9 holds.

Definition 3.2 (Infinite communication complexity). The *cost* of a protocol is the essential rank (see Definition 2.11) of the underlying tree. If a function $f: X \times Y \rightarrow Z$ has a communication protocol, the *communication complexity* $D(f)$ of f is the minimum cost of a protocol computing f .

3.1 Lower bounds

Just as in the finite case, we can use partition numbers to prove lower bounds on infinite communication problems:

Theorem 3.3. *If $f: X \times Y \rightarrow Z$ has a communication protocol, then $X \times Y$ can be partitioned into countably many f -monochromatic rectangles.*

Proof. Suppose f has a protocol P . Since P is a countable well-founded tree, it has countably many leaf nodes, each a finite distance from the root node. Thus the same argument as in the finite case tells us that the set of inputs reaching a given leaf node v form an f -monochromatic rectangle R_v . The collection of sets R_v for all leaf nodes v is our desired countable partition of $X \times Y$ by f -monochromatic rectangles. $\square$

Corollary 3.4. *The function $\text{EQ}: \Sigma^\omega \times \Sigma^\omega \rightarrow \Sigma$ defined by $\text{EQ}(x, y) = 1$ if $x = y$ and $\text{EQ}(x, y) = 0$ otherwise has no communication protocol.*

Proof. The set $\mathcal{F} = \{(x, x) : x \in \Sigma^\omega\}$ is uncountable, and by the same argument as in the finite case, any EQ-monochromatic rectangle contains at most one element of $\mathcal{F}$. This means that any EQ-monochromatic partition of $X \times Y$ requires uncountably many rectangles, and the contrapositive of Theorem 3.3 gives us the desired result. $\square$

Remark 3.5. If we define $\text{DISJ}: \Sigma^\omega \times \Sigma^\omega \rightarrow \Sigma$ defined by $\text{DISJ}(x, y) = 1$ if and only if $x_i \wedge y_i = 0$ for all $i \in \mathbb{N}$, then the fooling set $\mathcal{F} = \{(x, \bar{x}) : x \in \Sigma^\omega\}$ tells us that DISJ also has no communication protocol.

Remark 3.6. Infinite communication problems with communication protocols seem to correspond loosely with finite communication problems with *efficient* protocols (in the sense that the communication cost grows polylogarithmically with the input length). The functions EQ and DISJ give us examples of problems which are hard in both settings. The k -disjointness function $\text{DISJ}_{k,n}$ (for fixed k), which outputs 1 whenever x and y satisfy $\text{DISJ}_n(x, y) = 1$ and each contain at most k ones, has as infinite analog the *finite disjointness* function $\text{DISJ}_{<\omega}$ which outputs 1 whenever x and y satisfy $\text{DISJ}(x, y) = 1$ and contain finitely many ones. These functions give us an example of problems which are easy in both the finite and infinite setting.

The correspondence in the previous remark gives us heuristic evidence that Definition 3.1 is a reasonable extension of communication complexity to the infinite setting. The

next section provides more concrete evidence that our definition of infinite communication complexity is useful in descriptive set theory.

4 Infinite Karchmer-Wigderson games

In [6], Karchmer and Wigderson define for each function $f: \Sigma^n \rightarrow \Sigma$ a communication problem which completely characterizes the minimum (finite, fan-in 2) circuit depth of f . This communication problem has been named the *Karchmer-Wigderson (KW) game* and is a useful tool in proving circuit lower bounds. With our definition of infinite communication complexity and characterization of Borel sets with infinite circuits, we can prove connections between Borel sets and a certain infinite analog of KW games. For our purposes, it is best to phrase the game in terms of sets rather than a function.

Definition 4.1 (Karchmer-Wigderson games for sets). Fix two disjoint sets $A, B \subseteq \Sigma^\omega$. In the *KW game* $\text{KW}_{A,B}$, Alice is given a string $x \in A$, Bob is given a string $y \in B$, and they must output an index $i \in \mathbb{N}$ such that $x_i \neq y_i$ (such an index always exists since $x \neq y$).

Remark 4.2. The game $\text{KW}_{A,B}$ does not define a function, since there can be multiple indices $i \in \mathbb{N}$ for which $x_i \neq y_i$. Instead, we say that a protocol P solves the game $\text{KW}_{A,B}$ if the $P(x, y)$ -th bit of x and y disagree for all $x \in A, y \in B$.

Definition 4.3. We define the *communication complexity* $D(\text{KW}_{A,B})$ of a KW game $\text{KW}_{A,B}$ as the minimum cost of a protocol solving $\text{KW}_{A,B}$.

Definition 4.4 (Other KW games). We define for a set $S \subseteq \Sigma^\omega$ the KW game $\text{KW}_S = \text{KW}_{S, S^c}$. We define for any function $f: \Sigma^\omega \rightarrow \Sigma$ the KW game $\text{KW}_f = \text{KW}_{f^{-1}(\{1\}), f^{-1}(\{0\})}$.

Remark 4.5. When thinking about KW games, it is crucial to distinguish between always-terminating procedures and well-founded protocols. All KW games have always-terminating procedures, since Alice and Bob can first confirm that $(x, y) \in A \times B$ and then share the bits of x and y one-by-one until finding a bit where the two strings disagree. However, this procedure is usually ill-founded and thus not a protocol.

4.1 KW games and Borel separability

We now present the connection between KW games and Borel sets. This link is the motivation for our definitions of infinite communication complexity and KW games, and it is the essential piece of machinery for our upcoming results.

Definition 4.6. Suppose $A, B \subseteq \Sigma^\omega$ are disjoint. We say that a set $S \subseteq \Sigma^\omega$ *separates* A and B if $A \subseteq S$ and $S \cap B = \emptyset$. We call A and B *Borel separable* whenever A and B can be separated by a Borel set.

Theorem 4.7. *Two disjoint sets $A, B \subseteq \Sigma^\omega$ are Borel separable if and only if $\text{KW}_{A,B}$ has a communication protocol.*

Proof. First suppose that A and B are Borel separable. Then by Corollary 2.10, there exists a circuit C such that $C(x) = 1$ for all $x \in A$ and $C(y) = 0$ for all $y \in B$. Consider the following protocol for $\text{KW}_{A,B}$: Given $x \in A$ and $y \in B$, Alice and Bob begin by examining the output gate G of C , which satisfies $G(x) = 1$ and $G(y) = 0$. Let $G_1, G_2, \dots$ denote the children of G . If G is an $\vee$ -gate, then we have $G_i(x) = 1$ for some $i \in \mathbb{N}$ and $G_j(y) = 0$ for all $j \in \mathbb{N}$. Alice shares i with Bob in this case. If G is an $\wedge$ -gate, then we have $G_i(x) = 1$ for all $i \in \mathbb{N}$ and $G_j(y) = 0$ for some $j \in \mathbb{N}$. Bob shares j with Alice in this case. In both cases, Alice and Bob find a descendant G' of G such that $G'(x) = 1$ and $G'(y) = 0$. Alice and Bob repeat this process until finding a literal x_i or $\bar{x}_i$ evaluating to 1 on x and 0 on y . Thus Alice and Bob can guarantee correctness by outputting i . Since C is well-founded, this procedure is also well-founded and gives us a protocol for $\text{KW}_{A,B}$.

Now suppose $\text{KW}_{A,B}$ has a protocol. For each node v in P , let $R_v = A_v \times B_v \subseteq A \times B$ denote the set of inputs $(x, y) \in A \times B$ on which the protocol reaches v (the argument in Theorem 3.3 tells us that these sets are indeed rectangles). Suppose first that v is a leaf node. Letting i denote the label of v , the correctness of P tells us that $x_i \neq y_i$ for all $(x, y) \in R_v$. As R_v is a rectangle, this means that either $x_i = 1$ and $y_i = 0$ for all $(x, y) \in R_v$ or $x_i = 0$ and $y_i = 1$ for all $(x, y) \in R_v$. Call v a *positive leaf* if the first case holds and a *negative leaf* otherwise.

Now replace each of Alice's nodes with an $\vee$ -gate, each of Bob's nodes with an $\wedge$ -gate, each positive leaf labeled i with the literal x_i , and each negative leaf labeled i with the literal $\bar{x}_i$. The resulting circuit C is well-founded since P is a well-founded protocol. It therefore suffices to show that $C(x) = 1$ for all $x \in A$ and $C(y) = 0$ for all $y \in B$. To this end, let G_v denote the gate corresponding to the node v in P . We claim that, for each vertex v in P , we have $G_v(x) = 0$ for all $x \in A_v$ and $G_v(y) = 1$ for all $y \in B_v$. We saw already that the claim holds when v is a leaf node. If v is an internal node, we may inductively suppose that the claim holds for all children $v_1, v_2, \dots$ of v . If v is one of Alice's nodes, then for any $x \in A_v$, we have $x \in A_{v_i}$ for some $i \in \mathbb{N}$. Thus $G_{v_i}(x) = 1$ by the inductive hypothesis, implying that $G_v(x) = \bigvee_{i \in \mathbb{N}} G_{v_i}(x) = 1$. Similarly, if v is one of Bob's nodes, then every $y \in B_v$ satisfies $G_{v_i}(y) = 0$ for some $i \in \mathbb{N}$ and hence $G_v(y) = \bigwedge_{i \in \mathbb{N}} G_{v_i}(y) = 0$. Thus the claim holds for v , completing the induction. The root node v of P satisfies $R_v = A \times B$, so applying the claim to this node completes the proof. $\square$

Corollary 4.8. *A set S is Borel if and only if KW_S has a communication protocol.*

The conversion process in the previous proof exactly preserves the tree structure underlying the protocols and circuits involved. As the essential rank of this underlying tree is equal to both the cost of the protocol and the rank of the circuit, we obtain the following corollary:

Corollary 4.9. *If A and B are Borel separable subsets of Σ^ω , then the minimum rank of a Borel set separating A and B is precisely $D(\text{KW}_{A,B})$. The Borel rank of a Borel set B is precisely $D(\text{KW}_B)$.*

Remark 4.10. The definition of infinite communication complexity allows for the functions a_v and b_v to be arbitrary. Thus a protocol P solving some KW game $\text{KW}_{A,B}$ might not

have its nodes labeled by Borel functions. Given such a protocol, however, we can use the previous proof to obtain a circuit C separating A and B and then convert C back to a protocol P' solving $\text{KW}_{A,B}$ with Borel functions as labels. The underlying tree structure is preserved in the process. This means that the communication complexity of $\text{KW}_{A,B}$ is always “witnessed” by a protocol with the functions a_v and b_v Borel.

4.2 Monotone KW games

Definition 4.11. If $x, y \in \Sigma^\omega$, write $x \leq y$ whenever $x_i \leq y_i$ for all $i \in \mathbb{N}$. We call a function $f: \Sigma^\omega \rightarrow \Sigma$ *monotone* if $f(x) \leq f(y)$ whenever $x \leq y$. Call sets $A, B \subseteq \Sigma^\omega$ *monotone-separable* if there exists a monotone function $f: \Sigma^\omega \rightarrow \Sigma$ satisfying $f(x) = 1$ for all $x \in A$ and $f(y) = 0$ for all $y \in B$.

Definition 4.12. A circuit is *monotone* if it contains no negated literals $\bar{x}_i$ for $i \in \mathbb{N}$.

Because internal nodes of circuits are labeled with $\wedge$ - and $\vee$ -gates, circuits always compute monotone functions of the literals $0, 1, x_1, \bar{x}_1, x_2, \bar{x}_2, \dots$. Only the presence of negated literals $\bar{x}_1, \bar{x}_2, \dots$ allow circuits to compute functions which are not monotone. This means that every monotone circuit C computes a monotone function. These monotone circuits can also be characterized in terms of KW games.

Definition 4.13. Suppose $A, B \subseteq \Sigma^\omega$ are monotone-separable. In the *monotone KW game* $\text{mKW}_{A,B}$, Alice is given a string $x \in A$, Bob is given a string $y \in B$, and they must output an index $i \in \mathbb{N}$ such that $x_i = 1$ and $y_i = 0$ (monotone-separability implies that such an index must always exist).

Theorem 4.14. *Two monotone-separable sets A and B can be separated by a monotone circuit if and only if mKW_f has a communication protocol.*

Proof. If C is a monotone circuit separating A and B , then the protocol in the proof of Theorem 4.7 outputs $i \in \mathbb{N}$ such that some literal x_i or $\bar{x}_i$ in C evaluates to 1 on x and 0 on y . As C is monotone, this literal must be x_i , implying that $x_i = 1$ and $y_i = 0$. Conversely, if P is a protocol for $\text{mKW}_{A,B}$, then every leaf node v in P is a positive leaf. Thus the conversion process in Theorem 4.7 gives us a monotone circuit separating A and B . $\square$

5 Existence results using KW games

The rest of this paper is devoted to applications of infinite Karchmer-Wigderson games. In this section, we will prove positive results, demonstrating that Borel sets exist by solving the appropriate KW games. In particular, we will prove the analytic separation theorem and the fact that any monotone Borel function is computed by a monotone circuit.

5.1 The analytic separation theorem

The analytic separation theorem states that any two disjoint analytic sets are separable by a Borel set. In particular, if a set A and its complement are both analytic, then A must actually be a Borel set (this follows from taking $A' = \Sigma^\omega \setminus A$). Though the standard proofs of this theorem are not overly difficult, KW games give us a completely elementary and purely combinatorial proof:

Theorem 5.1. *Any two disjoint analytic sets $A, A' \subseteq \Sigma^\omega$ are Borel separable.*

Proof. By Theorem 4.7, it suffices to find a communication protocol for $\text{KW}_{A,A'}$. To this end, we use Theorem 2.12 to find nondeterministic CNFs $C(x, y)$ and $C'(x', y')$ computing A and A' , respectively. These circuits are shared knowledge between Alice and Bob, since they do not depend on x or x' . Let $G_1, G_2, \dots$ denote the clauses of C and $G'_1, G'_2, \dots$ denote the clauses of C' . Given $(x, x') \in A \times A'$, Alice and Bob first individually choose strings $y, y' \in \Sigma^\omega$ such that $C(x, y) = C'(x', y') = 1$. Alice then tells Bob a literal of G_1 which is satisfied, Bob tells Alice a literal of G'_1 which is satisfied, Alice shares a literal of G_2 which is satisfied, etc. If Alice and Bob ever share both the literals x_i and $\bar{x}'_i$ or both the literals $\bar{x}_i$ and x'_i , they output i .

This protocol always outputs a correct value, so we just need to show that it is well-founded. To this end, suppose for the sake of contradiction that the protocol has an infinite branch. Then this branch gives us a partial assignment to the bits of the strings x, x', y, y' such that $C(x, y) = C'(x', y') = 1$ and the partial assignments to x and x' are compatible. As C and C' compute monotone functions on their respective literals, any values x, x', y , and y' which agree with these partial assignments satisfy $C(x, y) = C'(x', y') = 1$. We choose an arbitrary extension such that $x = x'$. We then get $C(x, y) = C'(x, y') = 1$, implying that $x \in A \cap A'$. This contradicts the assumption that A and A' are disjoint. $\square$

5.2 Monotone Borel sets have monotone circuits

Every monotone circuit computes a monotone function, whether in the finite or infinite case. If a DNF computes a monotone function, then one can also obtain a monotone DNF for the same function by removing all negated literals. In the finite setting, every function has a (possibly exponentially large) DNF, and this means that a function is monotone if and only if it is computable by a monotone circuit.

The story changes when we only consider polynomial sized circuits. In [2], Tardos proved that there are monotone functions computable by polynomial-sized circuit families but not by polynomial-sized families of monotone circuits. This was both a surprising result and a disappointment to complexity theorists hoping to use superpolynomial lower bounds on monotone circuits to resolve the P vs. NP question.

Given the analogy between computability in the infinite setting and efficient computability in the finite setting (which we discuss in Remark 3.6 in the context of communication complexity and others discuss in [3], [9], [10], and [11]), one might expect a similar phe-

nomenon in the infinite setting. However, it turns out that all monotone Borel functions have monotone circuits. In other words, if a circuit computes a monotone function, one can find always find a monotone circuit computing the same function. This fact was first proven by Dyck (see [7]), and we provide an alternative proof using KW games:

Theorem 5.2. *Every monotone Borel function $f: \Sigma^\omega \rightarrow \Sigma$ is computed by a monotone circuit.*

Proof. It suffices to prove the stronger claim that any two monotone-separable analytic sets $A, A' \subseteq \Sigma^\omega$ are separable by a monotone circuit. By Theorem 4.14, it suffices to find a communication protocol for $\text{mKW}_{A,A'}$. We use the same game as in the proof of Theorem 5.1, except that Alice and Bob keep playing even if they share both the literals $\bar{x}_i$ and x'_i for some $i \in \mathbb{N}$ (i.e. the only stopping condition is sharing both x_i and $\bar{x}_i$).

We again see that this protocol always outputs a correct value, and to prove well-foundedness, suppose for the sake of contradiction that the protocol has an infinite branch. Then this branch gives us a partial assignment to the bits of the strings x, x', y, y' such that $C(x, y) = C'(x', y') = 1$ and such that $x_i \leq x'_i$ whenever x_i and x'_i are both assigned. We can extend this partial assignment to strings x, x', y, y' such that $C(x, y) = C'(x', y') = 1$ and $x \leq x'$. As $x \in A$ and $x' \in A'$, this contradicts the monotone-separability of A and A' . $\square$

6 Impossibility results using KW games

While the previous section focuses on positive results about Borel sets, this section focuses on negative results. We first discuss why techniques from finite communication complexity are insufficient to prove that a given KW game has no communication protocol. Nonetheless, we will prove that the KW games for infinite parity and ill-founded trees lack protocols and thereby present new, combinatorial proofs that these sets are not Borel.

6.1 Monochromatic partitions for KW games

Most lower bounds in finite communication complexity involve proving lower bounds on the number of f -monochromatic rectangles needed to partition a domain. We saw in Corollary 3.4 and Remark 3.5 that partition number lower bounds are useful in the infinite setting as well.

Aho, Ullman, and Yannakakis' results in [1] show that, for a boolean-valued function f with partition number $\chi(f)$, we have $D(f) \leq O((\log_2(\chi(f)))^2)$. Yannakakis extended these ideas in [12] to show that the same bound holds for all functions f , even those which are not boolean-valued. This means that, up to a quadratic loss, proving lower bounds on communication complexity in the finite setting amounts to proving lower bounds on partition numbers. Indeed, a wide array of lower bound techniques in communication complexity follow this approach.

One might expect that a similar result holds in the infinite setting. We proved that any communication protocol for f gives us a countable partition of $\Sigma^\omega \times \Sigma^\omega$ by f -monochromatic rectangles. Unfortunately, the converse fails dramatically. In particular, the existence of non-Borel functions tells us that there are KW games without communication protocols. Despite this fact, every KW game can be solved by a function f admitting a countable f -monochromatic partition of $\Sigma^\omega \times \Sigma^\omega$:

Theorem 6.1. *If $A, B \subseteq \Sigma^\omega$ are disjoint, then there is a function $f: \Sigma^\omega \times \Sigma^\omega \rightarrow \Sigma$ solving $\text{KW}_{A,B}$ such that $\Sigma^\omega \times \Sigma^\omega$ has a countable f -monochromatic partition by rectangles.*

Proof. We define $f(x, y) = 0$ when $x \notin A$ or $y \notin B$, and we set $f(x, y)$ to be the first index where x and y differ if $x \in A$ and $y \in B$. Then $\Sigma^\omega \times \Sigma^\omega$ is partitioned by the rectangles $A^c \times B$, $A \times B^c$, $A^c \times B^c$, and

$$(A \cap w b \Sigma^\omega) \times (B \cap w \bar{b} \Sigma^\omega)$$

for $w \in \Sigma^*$ and $b \in \Sigma$. □

Remark 6.2. This countable partition corresponds exactly to the always-terminating procedure for every KW game discussed in Remark 4.5. Thus in order to show that a given KW game lacks a communication protocol, we must take advantage of the fact that protocols are well-founded, not just always-terminating.

6.2 The Borel cut-and-choose game

The previous section tells us that, in order to obtain impossibility results using KW games, our techniques must go beyond counting monochromatic rectangles. We will now establish such a technique in the form of an infinite adversarial game.

Definition 6.3. The *Borel cut-and-choose game* $\text{BCG}_{A,B}$ takes as parameters two disjoint subsets A and B of Σ^ω . It is played between Player I (Cut) and Player II (Choose) as follows:

- We begin with $A^{(0)} = A$ and $B^{(0)} = B$.
- In the n -th round, Player I has to option to either partition $A^{(n-1)}$ or $B^{(n-1)}$ into countably many pieces $A_1^{(n-1)}, A_2^{(n-1)}, \dots$ or $B_1^{(n-1)}, B_2^{(n-1)}, \dots$.
 - If Player I partitioned A , Player II chooses some $j \in \mathbb{N}$ and we take $A^{(n)} = A_j^{(n-1)}$ and $B^{(n)} = B^{(n-1)}$.
 - If Player I partitioned B , Player II chooses some $k \in \mathbb{N}$ and we take $A^{(n)} = A^{(n-1)}$ and $B^{(n)} = B_k^{(n-1)}$.
- If there exist $i, n \in \mathbb{N}$ and $b \in \Sigma$ such that $x_i = b$ for all $x \in A^{(n)}$ and $y_i \neq b$ for all $y \in B^{(n)}$, then Player I wins. Otherwise, Player II wins.

This name is chosen based on resemblance of this game to those popularized by Jech in [4]. The Borel cut-and-choose game is open and therefore determined. Using KW games, we can show that this game provides another characterization of Borel separability.

Theorem 6.4. *Player I has a winning strategy in the game $\text{BCG}_{A,B}$ if and only if A and B are Borel separable.*

Proof. For the forwards direction, we prove the contrapositive. If A and B are not Borel separable, then Player II can always play so that $A^{(n)}$ and $B^{(n)}$ are not Borel separable for each $n \in \mathbb{N}$ (this is because if $\bigcup_{j \in \mathbb{N}} S_j$ and T are not Borel separable, then S_j and T are not Borel separable for some $j \in \mathbb{N}$). In particular, Player II can ensure that $A^{(n)}$ and $B^{(n)}$ are never separable by a set of the form $\{x \in \Sigma^\omega : x_i = b\}$ for $i \in \mathbb{N}$ and $b \in \Sigma$. Thus Player II has a winning strategy.

For the reverse direction, we use KW games. If A and B are Borel separable, then Theorem 4.7 gives us a communication protocol P for $\text{KW}_{A,B}$. Recall that for a node v in P , we define $R_v = A_v \times B_v$ to be the set of pairs $(x, y) \in A \times B$ such that the protocol reaches v on input (x, y) . Player I's strategy is to always ensure that $A^{(n)}$ and $B^{(n)}$ correspond to the sets $A_{v^{(n)}}$ and $B_{v^{(n)}}$ for some node $v^{(n)} \in P$ which is a distance of n from the root node. In particular, if $v^{(n-1)}$ is one of Alice's nodes with children $v_1, v_2, \dots$, Player I partitions $A^{(n-1)} = A_{v^{(n-1)}}$ into the sets $A_{v_1}, A_{v_2}, \dots$. If, on the other hand, $v^{(n-1)}$ is one of Bob's nodes with children $v_1, v_2, \dots$, Player I partitions $B^{(n-1)} = B_{v^{(n-1)}}$ into the sets $B_{v_1}, B_{v_2}, \dots$. We eventually reach a leaf node by the well-foundedness of P , at which point Player I wins. $\square$

We can also formulate a monotone version of Borel cut-and-choose games:

Definition 6.5. The *monotone Borel cut-and-choose game* $\text{mBCG}_{A,B}$ is defined identically to the non-monotone version, except that the winning conditions is changed: Player I wins if there exist $i, n \in \mathbb{N}$ such that $x_i = 1$ for all $x \in A^{(n)}$ and $y_i = 0$ for all $y \in B^{(n)}$, and Player II wins otherwise.

Theorem 6.6. *Player I has a winning strategy in the game $\text{mBCG}_{A,B}$ if and only if A and B are separable by a monotone Borel set.*

Proof. The forwards direction is proved identically to that of Theorem 6.4, using the fact that if $\bigcup_{j \in \mathbb{N}} S_j$ and T are not separable by a monotone Borel set, then S_j and T are not separable by a monotone Borel set for some $j \in \mathbb{N}$.

For the reverse direction, suppose A and B are separable by a monotone Borel set. Then by Theorem 5.2, there exists a monotone circuit separating A and B . Thus Theorem 4.14 gives us a protocol P for $\text{mKW}_{A,B}$, and the rest of the proof proceeds identically to that of Theorem 6.4. $\square$

The reverse directions of Theorems 6.4 and 6.6 give us a technique to prove that two sets A and B are not (monotone) Borel separable: providing a winning strategy for Player II in $\text{BCG}_{A,B}$ or $\text{mBCG}_{A,B}$. To obtain such a strategy, we will define a notion of “largeness” so

that Player II's strategy can simply be to pick a "large" set. For this to work, it must be the case that if $\bigcup_{n \in \mathbb{N}} S_n$ is large, so is some S_n . Baire category will give us one such notion of largeness.

6.3 A Baire-categorical lemma

In this section, we prove a lemma which will be helpful in the proofs that parity set and the set of ill-founded trees are not Borel. In this section, we let Γ denote either the binary alphabet Σ or the natural numbers $\mathbb{N}$. Given a string $v \in \Gamma^*$, we use the notation $[v]$ for the basic open set $v\Gamma^\omega$. For the following definitions, we use language suggestive of the combinatorial nature of the proofs in this section. The definitions are nonetheless standard and equivalent to the usual topological ones.

Definition 6.7. Call a set $A \subseteq \Gamma^\omega$ *nowhere dense* if for every $v \in \Gamma^*$, there exists $w \in \Gamma^*$ such that $[vw] \cap A = \emptyset$. A set is *meager* if it can be written as countable union of nowhere dense sets. A set is *nonmeager* if it is not meager and *nonmeager in $[v]$* if its intersection with $[v]$ is nonmeager. A set is *everywhere nonmeager in $[v]$* if its intersection with $[vw]$ is nonmeager for every $w \in \Gamma^*$.

It is immediate from the definitions that meager sets are closed under countable unions and hence that if $\bigcup_{n \in \mathbb{N}} A_n$ is nonmeager, so is some A_n . It is also immediate that if A is everywhere nonmeager in $[v]$, then it is also everywhere nonmeager in $[vw]$ for all $w \in \Gamma^*$. The Baire category theorem tells us that Γ^ω is everywhere nonmeager, and we give a brief combinatorial proof for completeness:

Lemma 6.8 (Baire category). *The set Γ^ω is everywhere nonmeager.*

Proof. Fix $v \in \Gamma^*$ and suppose the sets $A_n \subseteq \Gamma^\omega$ for $n \in \mathbb{N}$ are nowhere dense. Then since A_1 is nowhere dense, there exists $w_1 \in \Gamma^*$ such that $[v0w_1] \cap A_1 = \emptyset$. As A_2 is nowhere dense, there exists $w_2 \in \Gamma^*$ such that $[v0w_10w_2] \cap A_2 = \emptyset$. Repeating this process, we obtain a string $v0w_10w_20w_3 \dots \in \Gamma^\omega$ which does not lie in any A_n . Thus $\bigcup_{n \in \mathbb{N}} A_n \neq [v]$, as desired. $\square$

The following lemma is the key fact we need for our hardness results:

Lemma 6.9. *If $A \subseteq \Gamma^\omega$ is nonmeager in $[v]$, then there exists $w \in \Gamma^*$ such that A is everywhere nonmeager in $[vw]$.*

Proof. Suppose not. Then for every $w \in \Gamma^*$, there exists $z_w \in \Gamma^*$ such that $A \cap [vwz_w]$ is meager. Thus the set $A \cap \bigcup_{w \in \Gamma^*} [vwz_w]$ is meager. We also claim that the set $B = [v] \setminus \left(\bigcup_{w \in \Gamma^*} [vwz_w]\right)$ is nowhere dense. To this end, fix a string $s \in \Gamma^*$. If we can write s in the form vt for some $t \in \Gamma^*$, then we have $sz_t = vtz_t \in \bigcup_{w \in \Gamma^*} [vwz_w]$ and hence $sz_t \notin B$. Otherwise, we can choose $z \in \Gamma^*$ such that $sz \notin [v]$ and hence $sz \notin B$. This proves that B is nowhere dense, and we have in particular that $A \cap B$ is meager. Thus $A \cap [v]$ is meager as the union of the meager sets $A \cap \bigcup_{w \in \Gamma^*} [vwz_w]$ and $A \cap B$, contradicting our assumption. $\square$

6.4 Hardness of parity

A function $f: \Sigma^\omega \rightarrow \Sigma$ is a *parity function* if $f(x) \neq f(y)$ whenever x and y differ in a single bit. It is a classical result that no infinite parity function is Borel, and we are now ready to prove this fact using KW games:

Theorem 6.10. *No infinite parity function $f: \Sigma^\omega \rightarrow \Sigma$ is Borel.*

Proof. By Theorem 6.4, it suffices to exhibit a winning strategy for Player II in the game $\text{BCG}_{f^{-1}(\{1\}), f^{-1}(\{0\})}$. We claim that Player II can win by always maintaining the invariant that, for all $n \in \mathbb{N}$, the sets $A^{(n)}$ and $B^{(n)}$ are everywhere nonmeager in $[w^{(n)}]$ for some $w^{(n)} \in \Sigma^*$.

We first show that $A = f^{-1}(\{1\})$ and $B = f^{-1}(\{0\})$ are everywhere nonmeager in Σ^ω . To this end, fix $w \in \Sigma^*$. We have $wx = wx_1x_2 \cdots \in A$ if and only if $w\bar{x}_1x_2 \cdots \in B$ for all $x \in \Sigma^\omega$, and because the map sending $wx_1x_2 \cdots$ to $w\bar{x}_1x_2 \cdots$ is a category-preserving bijection from $A \cap [w]$ to $B \cap [w]$, we have that $A \cap [w]$ is meager if and only if $B \cap [w]$ is meager. As $[w]$ is nonmeager by Lemma 6.8, the only possibility is that A and B are nonmeager in $[w]$. This means that both sets are everywhere nonmeager.

We now show inductively that Player II's invariant can always be maintained, so suppose $A^{(n-1)}$ and $B^{(n-1)}$ are everywhere nonmeager in $[w^{(n-1)}]$. If $A^{(n-1)} = \bigcup_{j \in \mathbb{N}} A_j^{(n-1)}$, then some $A_j^{(n-1)}$ must be nonmeager in $[w^{(n-1)}]$. Thus Lemma 6.9 gives us a string $z \in \Sigma^*$ with $A_j^{(n-1)}$ everywhere nonmeager in $[w^{(n-1)}z]$. Player II chooses this j , and the invariant holds for $w^{(n)} = w^{(n-1)}z$. A symmetric argument tells us that this is also possible when Player I chooses to partition $B^{(n-1)}$.

Finally, fix $i, n \in \mathbb{N}$. We know that $A^{(n)}$ and $B^{(n)}$ are everywhere nonmeager in $[w^{(n)}]$ for some $w^{(n)} \in \Sigma^*$, and we extend $w^{(n)}$ to some string w of length at least i . As $A^{(n)}$ and $B^{(n)}$ are everywhere nonmeager in $[w]$, these sets both have nonempty intersection with $[w]$. Thus there exist strings $x \in A^{(n)}$ and $y \in B^{(n)}$ with $x_i = y_i$, so the winning condition for Player I does not hold. As $i, n \in \mathbb{N}$ were arbitrary, this strategy is winning for Player II. $\square$

6.5 Hardness of ill-founded trees

To prove that the set of ill-founded trees is not Borel, we need notions of largeness for ill-founded and well-founded trees which satisfy these properties. For ill-founded trees, we use a variant of being everywhere nonmeager. Let $\mathbb{N}^\omega$ denote the Baire space, and recall that trees are represented as prefix-closed subsets of $\mathbb{N}^*$.

Definition 6.11. For a collection A of ill-founded trees, write $X_A \subseteq \mathbb{N}^\omega$ to denote the set of strings x such that the ill-founded tree consisting precisely of the single infinite branch x lies in A .

This definition gives us a notion of largeness for ill-founded trees, since we can ask whether X_A is everywhere nonmeager in $[w]$ for a string $w \in \mathbb{N}^*$. For our notion of largeness

for well-founded trees, we use a property defined by Sipser in [10] (which he also uses to prove that the well-founded trees are not Borel, albeit in a different manner).

Definition 6.12. Let B be a collection of well-founded trees and $w \in \mathbb{N}^*$. Given a well-founded tree T , we define $wT = \{wx : x \in T\}$. We call B *large at w* if for every well-founded tree T , we have $S \cap [w] = wT$ for some tree $S \in B$.

Lemma 6.13. *If $B = \bigcup_{k \in \mathbb{N}} B_k$ is large at w , then we have for some $k \in \mathbb{N}$ that B_k is large at wk .*

Proof. Suppose not. Then for every $k \in \mathbb{N}$, there is a well-founded tree T_k such that $S \cap [wk] \neq wkT$ for all $S \in B_k$. The tree $T = \bigcup_{k \in \mathbb{N}} kT_k$ is well-founded. If we had $S \cap [w] = wT$ for any $S \in B$, then we would have $S \in B_k$ for some k and $S \cap [wk] = wkT$, contradicting the construction of T_k . Thus $S \cap [w] \neq wT$ for any $S \in B$. The fact that B is large at w then gives us our desired contradiction. $\square$

Note also that if B is large at w , then B is also large at wz for all $z \in \mathbb{N}^*$. With our two notions of largeness in hand, we are now ready to prove the main claim. We identify trees with elements of Σ^ω by fixing a bijection $\varphi : \mathbb{N} \rightarrow \mathbb{N}^*$ and mapping a tree T to the string $x \in \Sigma^\omega$ with $x_i = 1 \iff \varphi(x_i) \in T$.

Theorem 6.14. *The set of ill-founded trees is not Borel.*

Proof. Let A denote the set of ill-founded trees and B denote the set of well-founded trees. Because A is monotone, it suffices by Theorem 6.6 to exhibit a winning strategy in $\text{mBCG}_{A,B}$ for Player II. We claim that Player II can win by always maintaining the invariant that, for all $n \in \mathbb{N}$, there is a string $w^{(n)} \in \mathbb{N}^*$ such that $X_{A^{(n)}}$ is everywhere nonmeager in $[w^{(n)}]$ and $B^{(n)}$ is large at $w^{(n)}$.

We have that $X_A = \mathbb{N}^\omega$ is everywhere nonmeager and that B is large at the empty string ε . To see that Player II can maintain the invariant, we can therefore use induction. Suppose $X_{A^{(n-1)}}$ is everywhere nonmeager in $[w^{(n-1)}]$ and $B^{(n-1)}$ is large at $w^{(n-1)}$ for some string $w^{(n-1)} \in \mathbb{N}^*$. If $A^{(n-1)} = \bigcup_{j \in \mathbb{N}} A_j^{(n-1)}$, then Lemma 6.9 tells us that some $X_{A_j^{(n-1)}}$ is everywhere nonmeager in $[w^{(n-1)}z]$ for some $z \in \mathbb{N}^*$. Thus Player II can choose j , and the invariant holds for $w^{(n)} = w^{(n-1)}z$. If, on the other hand, Player I partitions $B^{(n-1)} = \bigcup_{k \in \mathbb{N}} B_k^{(n-1)}$, then Lemma 6.13 tells us that some $B_k^{(n-1)}$ is large at $w^{(n-1)}k$. Thus Player II can choose k , and the invariant holds for $w^{(n)} = w^{(n-1)}k$.

To show that this strategy is winning for Player II, it suffices to prove that for any $n \in \mathbb{N}$ and string $v \in \mathbb{N}^*$, we have either $v \notin S$ for some $S \in A^{(n)}$ or $v \in T$ for some $T \in B^{(n)}$. Indeed, if v and $w^{(n)}$ are compatible, then the largeness of $B^{(n)}$ at $w^{(n)}$ gives us a tree $T \in B^{(n)}$ with $v \in T$. If v and $w^{(n)}$ are incompatible, then we can choose $x \in X_{A^{(n)}}$ with v and x incompatible. The tree S consisting precisely of the infinite branch x satisfies $S \in A^{(n)}$ and $v \notin S$, as desired. $\square$

Remark 6.15. This proof actually tells us that no monotone Borel set separates the ill-founded trees consisting precisely of a single infinite branch from the well-founded trees.

Since the monotone closure of the trees consisting of a single infinite branch is the set of all ill-founded trees, this result is no stronger than the fact that the ill-founded trees do not form a Borel set.

7 Thoughts on finite circuits

As the set of ill-founded trees is analytic (one can nondeterministically guess the branch), we have proven the following using KW games (in Theorem 5.1 and Theorem 6.14):

1. A set is Borel if and only if it is simultaneously analytic and co-analytic.
2. The classes of Borel sets, analytic sets, and co-analytic sets are all distinct.

These facts have corresponding statements for finite circuits. The first statement corresponds to the claim $P/\text{poly} = NP/\text{poly} \cap \text{coNP}/\text{poly}$, and the second corresponds to the proposition that $P/\text{poly} \neq NP/\text{poly} \neq \text{coNP}/\text{poly}$ (which implies $P \neq NP \neq \text{coNP}$). Adapting the arguments in this paper to the finite case provide a potential route to resolving the P vs. NP and NP vs. coNP problems. However, there are many differences between the finite and infinite settings which suggest that such a task will be difficult:

- The claim $P/\text{poly} = NP/\text{poly} \cap \text{coNP}/\text{poly}$ is generally believed to be false, while the infinite version of this claim is true.
- Monotone Borel functions have monotone circuits (Theorem 5.2), while monotone functions in P do not all have polynomial-sized monotone circuit families ([2]).
- $D(f)$ and $\log_2(\chi(f))$ are quadratically related in finite communication complexity, while all infinite KW games (even those without protocols) have countable partitions (Theorem 6.1).

It seems that well-foundedness, although related to “efficiency,” is not a perfect analog. While “efficient” finite procedures almost always correspond to well-founded infinite procedures, many well-founded procedures have inefficient finite counterparts. In some sense, this observation tells us that P is weaker relative to NP than the Borel sets are to the analytic sets. This explains some of the aforementioned differences between the finite and infinite setting and provides heuristic evidence that proving $P \neq NP$ using ideas from descriptive set theory may still be a promising approach.

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